



Yale College
Science & Quantitative Reasoning

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Simulating In-Memory Probabilistic Neural Networks Using P-Bit Constructs

Probabilistic computing is an emerging paradigm that addresses the memory bottlenecks in traditional von Neumann architectures. A recently proposed building block for this approach is the probabilistic bit, or P-bit, which flips between binary states with a controllable probability. At the hardware level, P-bits can be realized using voltage-controlled magnetic tunnel junctions (VCMA-MTJs), allowing the same device to serve as both memory and a source of true randomness. Such devices are attractive for compute-in-memory architectures, especially in neural networks that benefit from stochasticity such as Bayesian inference or diffusion-based generative models.

My initial research goal was to explore how P-bit inspired architectures could be studied without direct access to nanoscale devices. To that end, we built a Verilog-based simulation environment to model P-bits digitally and integrate them into a minimal convolutional neural network (CNN)-like pipeline. The project expanded over the summer from modeling a single P-bit to developing full stochastic-to-binary conversion modules and running cycle-accurate testbenches for CNN-style inference.

My hypothesis was that in-memory probabilistic elements could support neural computations with reduced data movement and circuit complexity while maintaining statistical reliability.

The first stage of the project implemented a parameterizable P-bit (PBIT.v), which stores a bit but flips probabilistically on write. Flip rates are controlled by a lookup table indexed by a four-bit control word. A simple random number generator provided the stochastic draw for toggling. We then built supporting modules (BSConverter.v) to translate between binary numbers and stochastic bitstreams, enabling conventional data to be interfaced with probabilistic logic. Finally, I composed these modules into a small CNN-like pipeline (StochasticCNN.v) with a convolution pass and a fully connected accumulation stage.

Simulation results showed that P-bits flipped at frequencies consistent with their programmed probabilities, confirming controllable stochastic behavior. The binary-stochastic conversion

modules produced expected averages, with errors decreasing as bitstream length increased. The stochastic CNN produced stable accumulated outputs after multiple cycles, demonstrating that in-situ randomness can effectively drive neural computation. Importantly, the architecture collapsed the traditional generate–sample–transfer–compute sequence into a more efficient generate–compute scheme.

The results suggest that even in simplified digital simulation, probabilistic memory elements can approximate neural inference while avoiding external sampling overhead. This opens a path for extending the design to more complex networks and, ultimately, mapping it to FPGA or ASIC implementations for power and timing evaluation. Future directions include scaling to multilayer networks, incorporating realistic device-level noise models, and exploring hybrid probabilistic-deterministic architectures.

My research experience also centered on developing design discipline. I learned to structure Verilog modules systematically, to use seeded randomness for reproducible stochastic experiments, and to debug nondeterministic timing errors. I also gained experience designing testbenches that probe both functionality and statistical properties. More broadly, I learned how to frame hypotheses at the architectural level, design experiments to isolate specific effects, and document results for reproducibility. This project strengthened my understanding of how hardware concepts translate into system-level behaviors, deepened my interest in probabilistic AI hardware, and gave me confidence to pursue further work in accelerator architecture at Yale with Professor Linghao Song.

There were challenges, including resolving race conditions caused by mixing blocking and non-blocking assignments, and mitigating periodicity in random number generation. Addressing these hurdles improved my debugging skills and taught me to anticipate how stochastic systems require both statistical and logical validation. Having the time and support to focus on this project was invaluable. It directly shaped my career trajectory toward AI accelerators and reinforced my desire to continue research in STEM.

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